Epistatic Circuits
Walsh–Hadamard Interaction Decomposition for Transformer Circuit Discovery

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The Problem

Circuit discovery methods report each component's **marginal** contribution—the change in a metric when one head is ablated in isolation. A head whose effect depends on the state of another head has no marginal single score: its contribution is a function of the circuit it sits inside.

IOI and RTI circuits carry **13–15%** of their total variance at order 2 (pairwise interactions). Gendered pronoun resolution carries $\sim$1%. An additive summary discards everything above order 1.

How much interaction structure does each discovery method preserve, and what does that structure contain?

Walsh–Hadamard Decomposition

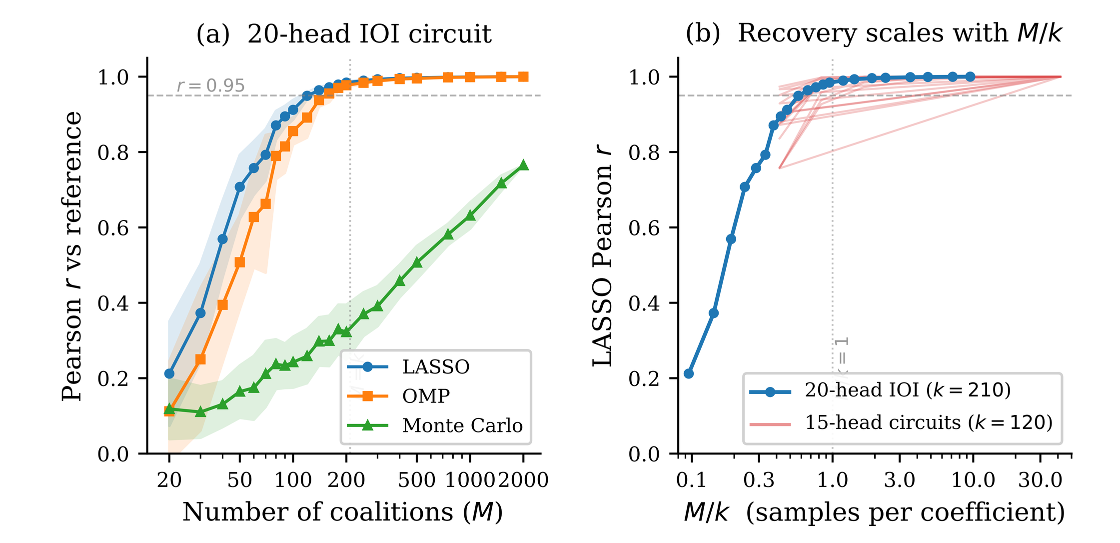
For n heads, the 2^n coalition function $v(S)$ (metric value when only heads in S are active) has a unique decomposition into Walsh coefficients w_T for each subset $T \subseteq [n]$:

$$v(S) = \sum_{T \subseteq [n]} w_T \prod_{i \in T} (-1)^{\mathbb{1}[i \notin S]}$$

$|T| = 1$: individual head effects. $|T| = 2$: pairwise interactions. The energy w_T^2 at each order sums to the total variance.

Compressed sensing recovers the spectrum from far fewer than 2^n coalitions. LASSO on $M = 140$ random coalitions recovers the full pairwise spectrum of a 15-head circuit at $r = 0.96 \pm 0.01$ (split-half $r = 0.997$). This is a **7,490$\times$ compression** over the exhaustive $2^{15} = 32{,}768$ sweep.

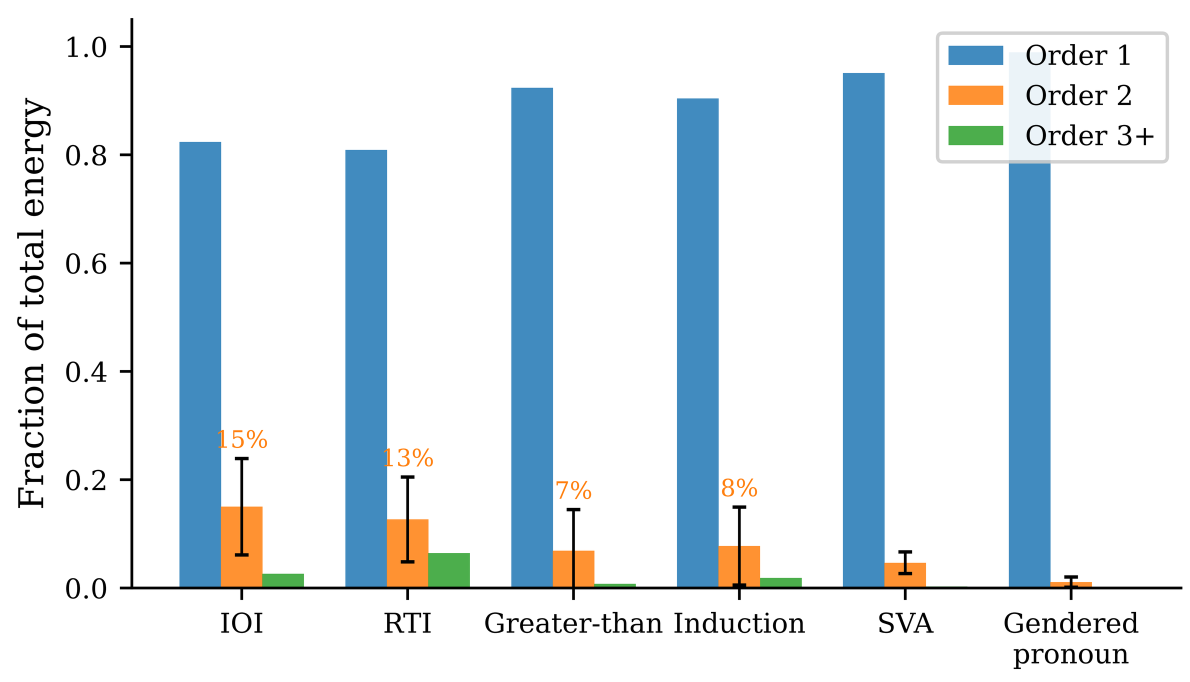
Sparse Recovery



Method	$M = 140$	$M = 500$
LASSO	0.96 ± 0.01	0.99 ± 0.00
OMP	0.91 ± 0.04	0.98 ± 0.01
Monte Carlo	0.78 ± 0.03	0.91 ± 0.01

Recovery correlation (r) between sparse estimate and ground truth from exhaustive sweep. LASSO dominates at low M.

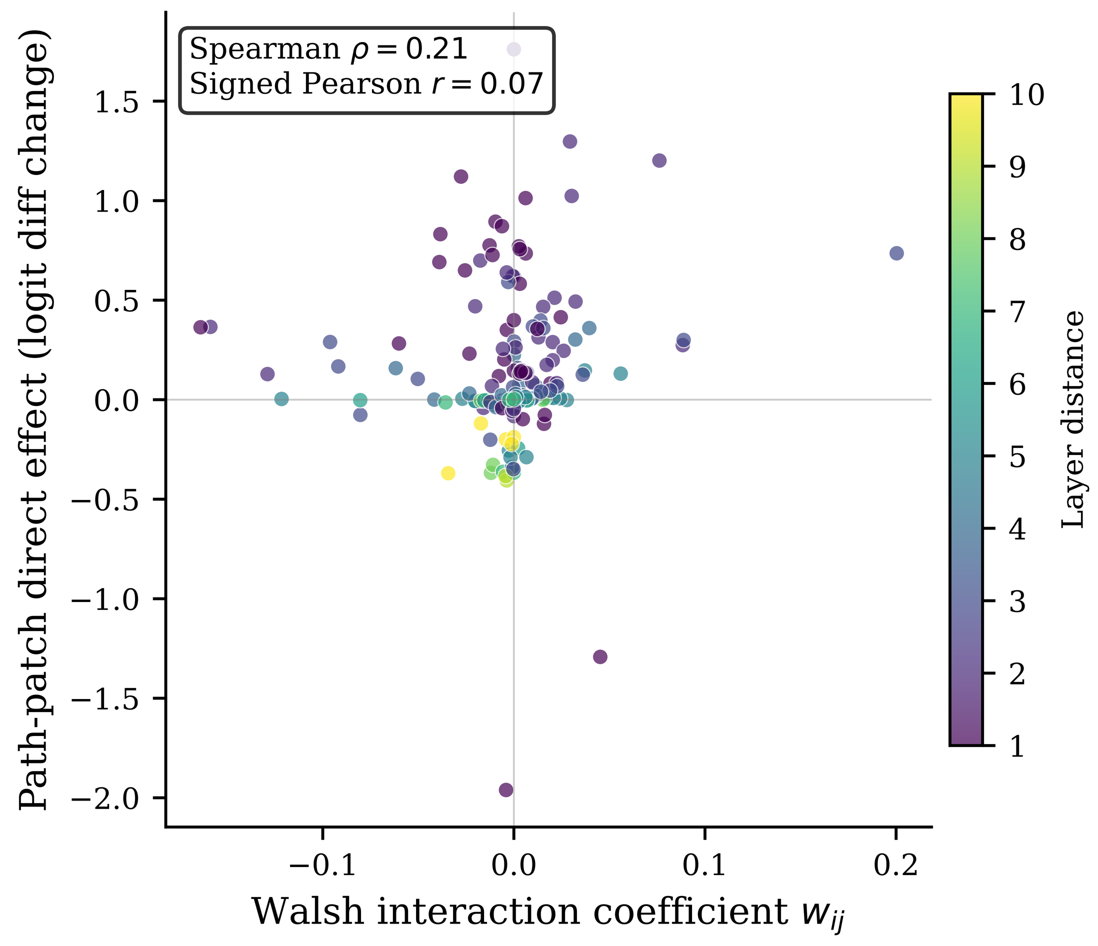
Energy Spectrum Across Six Tasks



Task	Order 1	Order 2	Order 3+
IOI	0.82	0.15	0.03
RTI	0.81	0.13	0.06
Greater-than	0.92	0.07	0.01
Induction	0.90	0.08	0.02
SVA	0.95	0.05	0.00
Gendered pronoun	0.99	0.01	0.00
Random	0.92	0.06	0.02

60 sweeps across 6 tasks in GPT-2 small. Tasks requiring coordinated attention routing (IOI, RTI) carry 13–15% pairwise energy. Random head sets carry $\sim$6%, confirming that interaction structure reflects functional coupling.

Discovery Methods Disagree



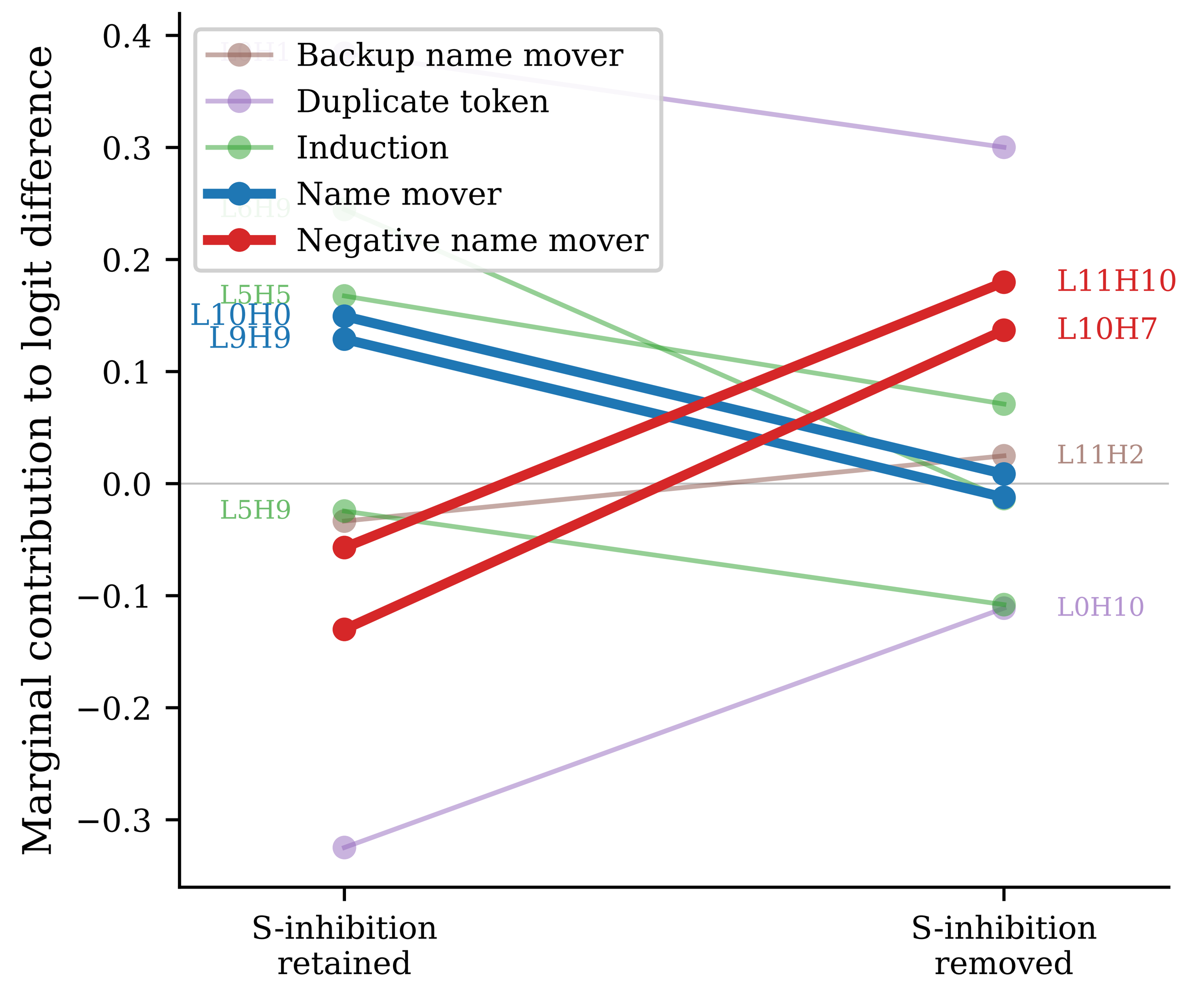
Comparison	Spearman ρ	Pearson r
Walsh vs EAP-IG	0.58	0.73
Walsh vs path patching	0.21	0.07

EAP-IG captures moderate interaction structure ($\rho = 0.58$). Path patching is **nearly orthogonal** ($r = 0.07$): its top-ranked pair (L10H7→L11H1) ranks 42nd by Walsh.

Weight geometry predicts nothing. Cross-validated $R^2 < 0$ across all 30 conditions (zero, mean, resample ablation $\times$ 10 geometric primitives). 32% of pairwise geometric predictions have the wrong sign.

What the Interactions Contain: Complementation Test

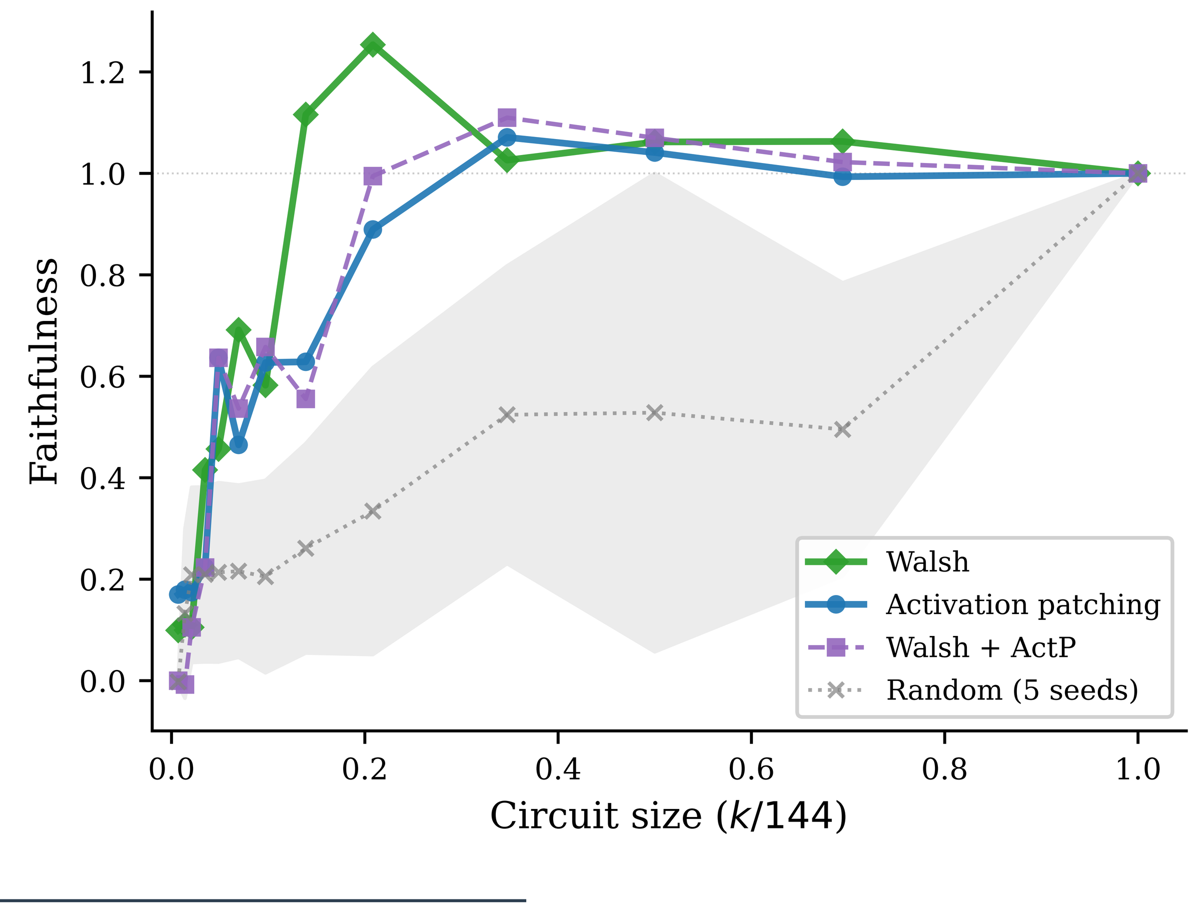
S-inhibition heads and negative name movers merge into one functional unit in 1000/1000 bootstrap resamples (ARI = 0.571 against published IOI taxonomy). The merge holds across ablation methods, circuit selection methods, and linkage criteria.



Negative name movers **invert sign** when S-inhibition is removed: L10H7 goes from -0.130 to $+0.137$, L11H10 from -0.057 to $+0.180$ (both flip in 1000/1000 draws). Name movers decay toward zero without inverting.

An additive summary reports negative name movers as harmful. Their actual effect depends on S-inhibition state.

Walsh-Ranked Circuits Are Faithful (MIB)



Method	CPR ↑	CMD ↓
Walsh	0.996	0.100
Activation patching	0.937	0.095
Walsh + activation patching	0.956	0.104
Random	0.608 ± 0.194	0.402 ± 0.180

Walsh reaches $f > 1.0$ at $k = 20$ heads (14% of model), where activation patching is at $f = 0.63$. All 144 heads in GPT-2 small ranked; 16 of top 20 overlap between methods. Combining adds noise rather than complementary information.

Takeaway

A head's measured contribution is a property of the head together with the background it was measured against.